

The AI browser blind spot

What Comet, Brave Leo, Claude in Chrome, ChatGPT and Gemini leave on disk, and how much of it can say who acted.

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PREVIOUSLY

- DART India Founding Engineer, **Rippling**
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- Cyber Security Engineer, **Microsoft**
- Lead MDR & Threat Hunter, **IBM**

Someone pasted a confidential contract into an AI browser. We read it back off the disk.

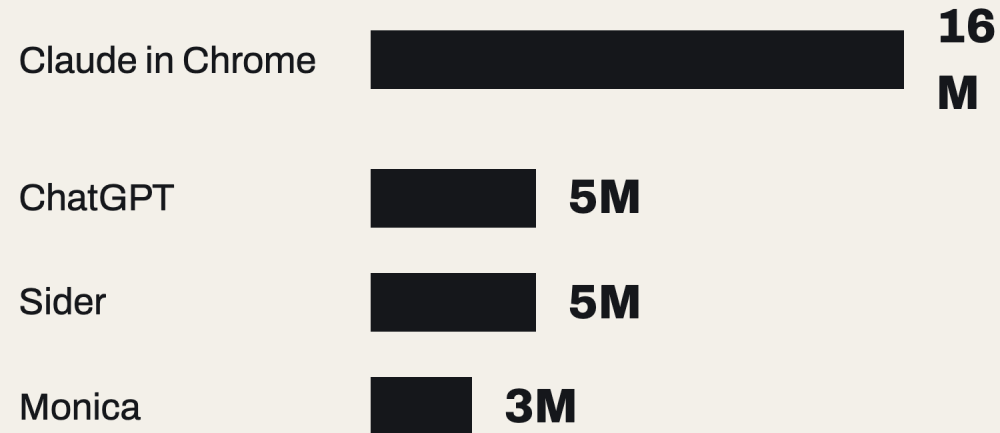
```
$ ccl_chromium_reader 'Profile 1/IndexedDB/https_www.perplexity.ai_0'  
store: keyval-store (the Comet assistant's local cache)  
step_type: INITIAL_QUERY  
query: summarize this: CONFIDENTIAL, INTERNAL ONLY (fictional lab document)  
  Document ID: COCON-CANARY-COMET-03  
  Subject: Ostrich vendor contract, negotiation position  
  1. Walk-away price is 1.8M. Opening offer 1.2M.  
  2. Known weakness: their SLA on data residency is unenforceable.  
  ... 2 more captured lines not shown
```

Real capture · lab Comet profile collected 20 Sep 2026, parsed 26 Sep 2026

Would your collection have picked this up? Could you prove a person pasted it, and not an agent?

Treat them as a new class of LOLBins

Chrome Web Store users [4]



1 in 6 enterprise users has an AI extension [3]



DLP blind spot

A paste into an assistant creates no file.



Dec 2025 · 900k+ users

Fake AI sidebars shipped chats to attackers every 30 min [1]



Aug 2026 · 6 stealer families

claude.ai sessions stolen; Anthropic signed users out [2]

AI-assisted browsing comes in four variants.



AI-native browser

Comet · Brave Leo

Own profile folder



Built into the browser

Gemini in Chrome [9]

Hidden storage partition



Agent extension

Claude in Chrome [8] ·
ChatGPT extension [7]

Extension store +
desktop bridge



Desktop agent driving a browser

Claude Code · Codex ·
browser-use MCPs

App data + OS grants



An endpoint can carry an AI agent without any AI browser installed.

AI browsers open two blind spots in browser forensics.



Coverage

Collection misses where AI browsers write.

Fixable: collect more



Attribution

Person or agent? The rows look alike.

Managed, not solved

ACT 02 OF 04

On disk

What the network misses, and what each AI product actually wrote, from our own lab captures.

They all look like Chrome in logs and fingerprints. On the endpoint they register themselves for bridging

🌐 On the wire

Comet

UA: Chrome

Atlas

UA: Chrome

Chrome

UA: Chrome

1/7

agents stopped by free
cloudflare AI-bot
blocking [5]

7/7

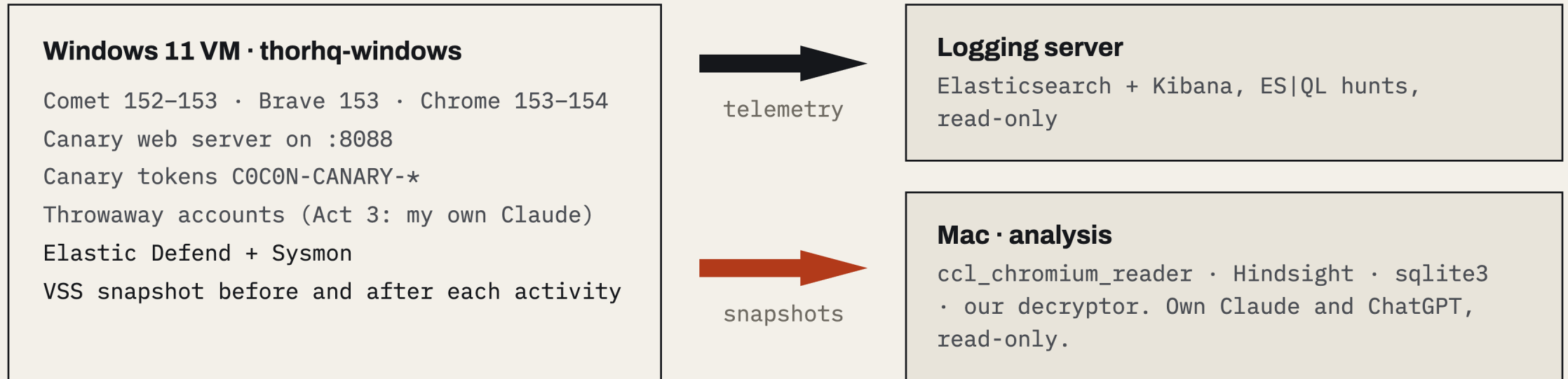
caught by page
behaviour, which only
the website sees

🔍 On the endpoint • macOS

```
$ cd ~/Library/Application Support/Google/Chrome
$ ls NativeMessagingHosts/ | grep -E
'anthropic|openai|perplexity'
com.anthropic.claude_browser_extension.json
com.anthropic.claude_code_browser_extension.json
com.openai.codexextension.json
com.perplexity.comet.json
```

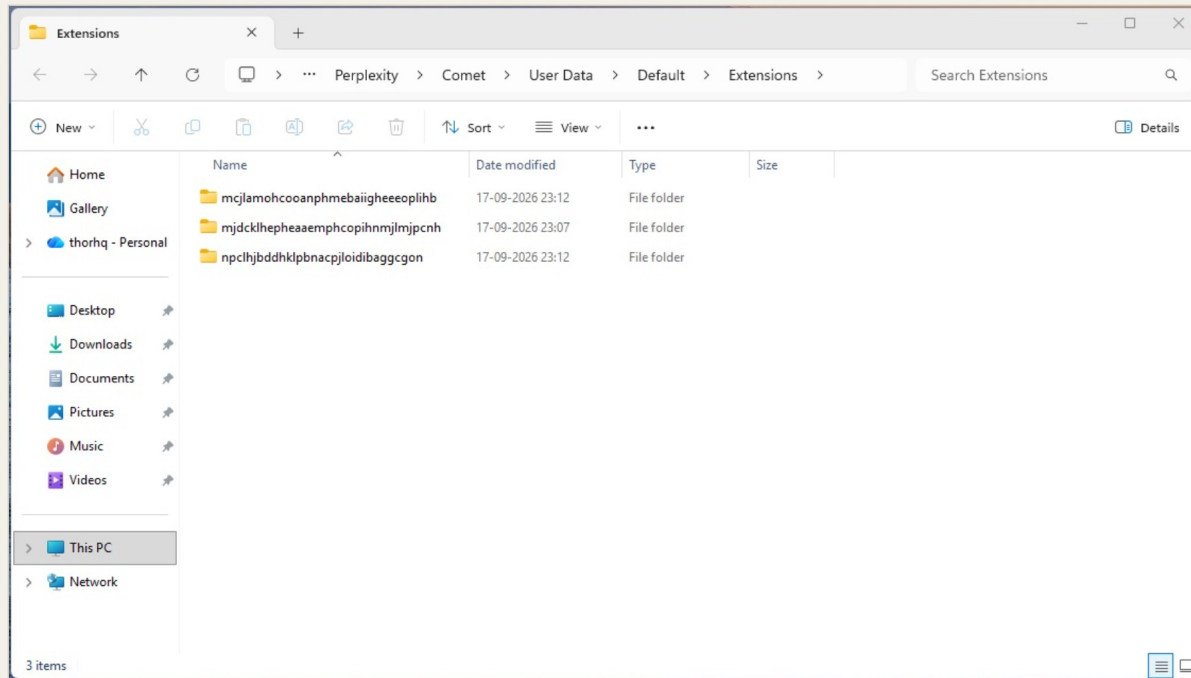
Real capture • 26 Sep 2026 • each file registers a browser-to-desktop-app bridge

Every artifact claim in this act comes from this lab.



Canary strings, real use of each product, disk snapshots before and after. One Windows VM: one run per product for artifacts, three runs per driver for attribution. Plus my own Mac, read-only. What we observed, not a survey.

Comet adds three extensions to the profile on first run. The user is not given a choice.



Real screenshot · lab VM · 20 Sep 2026

**comet-agent**
debugger · nativeMessaging · all sites

**Comet**
the assistant

**Comet Web Resources**
the assistant's UI

Secure Preferences · from_webstore: true
Recorded as Web Store installs. They came with the installer.

On disk · bundled extensions

Perplexity keeps chats on its servers, But chat prompts are still on the laptop.

Comet · IndexedDB

```
$ ccl_chromium_reader .../https_www.perplexity.ai_0
query_str = Remember that my project codename is C0C0-CANARY-COMET-02
thread_title = Remember that my project codename is C0C0-CANARY-COMET-02
backend_uuid = 3e3e71d3-4cdc-4e5b-ad7d-157b9b210a4b
```

Real capture · the thread ID points to the cloud record

Comet · History, UTC

```
$ sqlite3 'Profile 1/History'
07:52:43 /canary-COMET-01.html
07:55:18 /secret-COMET-03.txt
07:55:57 /Form.html
07:56:53 /submit
```

Real capture · 20 Sep 2026

Preserve the thread ID with account and time data. It helps the vendor find the cloud record.

Brave encrypts the chat text, but not the timeline.

Brave Leo · AIChat (SQLite) · 20 Sep, UTC

```
$ sqlite3 '.../User Data/Default/AIChat'
```

time	who	model	text
07:59:55	human		76 31 30 ...
08:00:01	assistant	qwen	76 31 30 ...
08:01:00	human		76 31 30 ...
08:01:02	assistant	qwen	76 31 30 ...

Real capture · "76 31 30" is ASCII v10: encrypted with the user's key · model is chat-qwen

Decrypt, live on the Windows VM

```
> python chromium_decrypt.py "...\\Brave-Browser\\User Data" Default
app_bound_encrypted_key present: False
v10 AES key recovered via user-DPAPI: 32 bytes
HUMAN 'What is COCON-CANARY-BRAVE-01?'
ASSISTANT 'I don't have specific information about ...'
HUMAN 'summarize http://192.168.1.14:8088/canary-BRAVE-01.html'
ASSISTANT 'I cannot access local network addresses ...'
title 'COCON CANARY BRAVE Device Information'
```

Live capture · thorhq-windows, as the user · 26 Sep 2026

Gemini in Chrome keeps its own cookies in a hidden partition.

It is built into Chrome, not an extension [9].

Chrome profile · macOS

```
$ ls ~/Library/Application Support/Google/Chrome/Default/Storage/ext/glic/  
66A834677761
```

```
$ echo -n glicpart | shasum -a 256 | cut -c1-12  
66A834677761
```

```
$ ls ../glic/66A834677761/
```

Cookies · Local Storage · Session Storage · Network Persistent State · GPUCache ...

The folder name is derived from the partition name

Browser monitoring Tools that only read Default/Cookies and Default/History never open this folder.

Quick Fix : Add Storage/ext/*/ to your collection path

Claude in Chrome records every site you "always allow" its agent on, with timestamps.

Claude in Chrome · extension storage

```
$ ccl_leveladb 'Local Extension Settings/fcoeoab...'
  key=permissionStorage
{ "permissions": [
  { "action": "allow", "duration": "always",
    "scope": { "type": "netloc", "netloc": "<domain>" },
    "createdAt": 1775908040717,
    "lastUsed": 1775908048950 },
  ... 6 more grants ] }
```

Real capture · domains redacted



Per-site timeline

createdAt, lastUsed. Leads, not proof.



Unsigned JSON

Editable to pre-approve sites [\[16\]](#)



Login tokens: elsewhere

v1.0.94 keeps them in Local Storage, not this store



"Allow this time"

Not stored. Only "always".

18,848

"Is the desktop app connected?"
checks, written by OpenAI's
ChatGPT extension.

ChatGPT extension · extension storage

```
$ ccl_levelldb 'Local Extension Settings/hehggadaopoacedllhhajmbjkdcmajg'  
key NATIVE_HOST_STATUS · records: 18,848 (live + superseded)  
hostName: com.openai.codexextension  
lastChecked, first to last (UTC): 2026-05-26 00:54:25 to 2026-07-06 07:55:57
```

Real capture · own Mac, read-only · no chat or login data in this store

The chats live on OpenAI's servers. This store still gives you 40 days of timestamps with the desktop app recorded as connected.

macOS records what each AI agent was allowed to do, even after the app is gone.

~/Library/Application Support/com.apple.TCC/TCC.db · AI apps, allowed

app	granted
com.anthropic.claude-code	AppleEvents, Desktop, Documents, Downloads
com.anthropic.claudefordesktop	Desktop, Documents, Downloads
com.openai.codex	AppleEvents, Desktop, Documents, Downloads
ai.perplexity.comet	Bluetooth, WebAuthn
com.openai.atlas	Microphone, Bluetooth, Downloads, WebAuthn

Real capture · own Mac, read-only · summarised per app

Atlas app	Gone from /Applications. Still present: ~/Library/LaunchAgents/com.openai.atlas.update-helper.plist and 4 permission grants.
------------------	--

Forensic artifacts left on disk

Comet • Win/mac	Brave Leo • Win	Claude ext • mac	ChatGPT ext • mac	
Found	Found	Not tested	Server only	Prompt or chat text
None	AIChat	None	Empty	Dedicated chat database
History	Timestamps	Grant lastUsed	40 days of Heartbeat	Activity timeline
3 bundled	None	None	None	Bundled hidden extensions
mac yes, Win no	None	Found	Found	Native-messaging bridge
v20 Win, Keychain mac	v10 decrypted	Key names only	Key names only	Session or credentials

Found on disk Encrypted, partial or OS-dependent Server-side only None or not tested

ACT 03 OF 04

Agent or Human?

No single layer proves human or agent. Browser artifacts, app data, process telemetry and identity have to be lined up by time.

Process telemetry shows how the browser was launched, not who drove it later.

👤 **Person**

explorer.exe



comet.exe

desktop launch · Elastic, 20 Sep

</> **Script**

sshd.exe



cmd.exe



comet.exe --remote-debugging-port=9222



Agent in an open browser

comet.exe, already running



no new process

form submitted 07:56, 20 Sep

Hunt scripted launches here. An agent inside an open browser starts nothing to see.

Same form, four drivers. History catches two of them and misses one.

```
Open http://thorhq-windows:8088/form-hva.html and fill the form:  
name COCON-CANARY-HVA-CA-T1-1, email hva-a-1@lab.local, notes  
'Phase 3 agent run 1'. Then submit it.
```



Me, typing

TYPED ·
FROM_ADDRESS_BAR

New tab · filled in 19–26 s

**Baseline: typed in the
address bar**



Claude in Chrome

LINK · FROM_API

Tab group "Claude" · 11–14 s

**Caught by FROM_API +
"Claude" tab group**



Playwright script

TYPED · FROM_API

New tab · filled in 2–6 s

**Caught by FROM_API
in History**



Comet's own agent

TYPED ·
FROM_ADDRESS_BAR

My tab · filled in 55 s

**Missed: identical to
address-bar typing**

Hunt: History visits with the FROM_API bit set (0x08000000).

Who drove · four tests

Process flags scripts and desktop bridges. History flags some agents.

👤 Person	💬 Assistant chat	⚡ Agent mode	</> Script	
explorer.exe	Blind: nothing new	Blind: nothing new	Flags: cmd.exe parent	Process launch
Renderers only	Partial: desktop bridge host	Partial: bridge host; renderers as a person	Renderers only	Child processes
none	Blind: none	Blind: none	Flags: debugging port	Launch flags
Chrome	Blind: Chrome	Blind: Chrome	Blind: Chrome	User-agent, TLS
FROM_ADDRESS_BARR	Partial: prompt cache	Partial: Claude FROM_API, Comet none	Flags: FROM_API	History on disk

■ Flags it on its own ■ Partial or product-dependent □ Blind, or the person baseline

For cloud-backed assistants, the vendor may keep more than the endpoint does.



In our lab: Comet caches **chat** prompts locally (slide 11), but an **agent-mode task** prompt had 0 hits on disk. Only its result was there (History, autofill).

Perplexity / Comet [11]

Until deleted

Threads. Assistant page context up to 30 days. Incognito 24 h.

Google Gemini [12]

18 months

By default. 72 h even with activity off.

Microsoft Copilot [13]

18 months

By default.



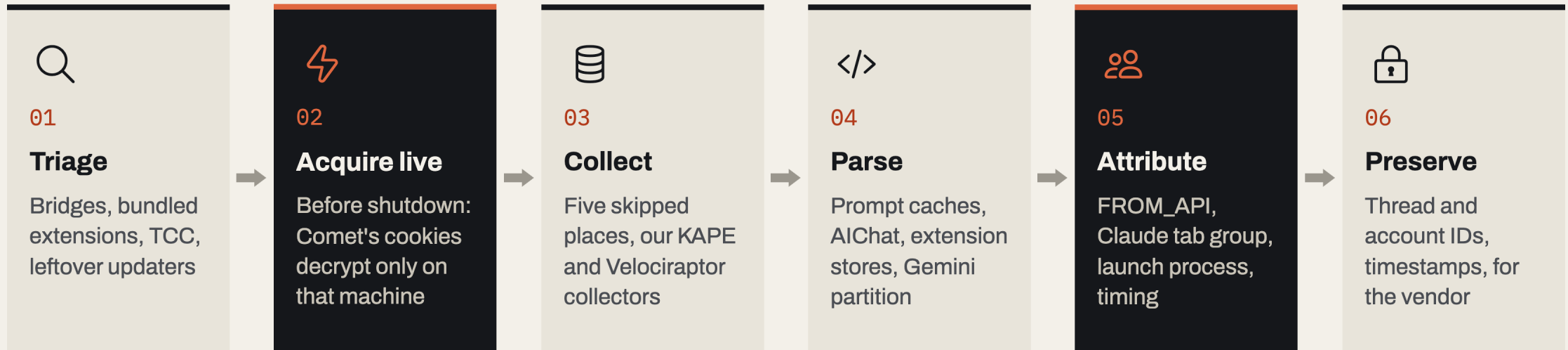
Preserve thread IDs, account IDs and timestamps for a vendor request.

ACT 04 OF 04

Close the gap

How to work a case, where to collect, the collectors we wrote, and the hunts to start with.

Investigating an AI-browser case, step by step



Five places standard collection skips, and what collects them

Place	What's there	KAPE	Velociraptor
 AI profile stores	Comet prompt cache, Brave AIChat: prompts, thread IDs	Yes	Yes
 Hidden partitions	Chrome Storage/ext/glic: Gemini's own storage	Comet only	Yes
 Extension stores	Claude grants, ChatGPT heartbeat, Comet's bundled agents	Comet only	Yes
 Bridges	Native-messaging hosts: which desktop programs can drive the browser	No	Yes
 OS and installer	Comet updater log; on macOS, TCC grants and LaunchAgents	Updater log	Updater log

KAPE had 31 browser targets, none for Comet, Atlas or Dia, so I wrote two, plus a **Velociraptor** artifact and an extractor.

Hunt leads, by confidence. These ones are yours to write.



Observed · Simulated

comet.exe started by cmd.exe with --remote-debugging-port



Observed · Simulated

comet.exe plus the WOW6432Node\Perplexity\Update key



Observed · lab · new

History visits with FROM_API (0x08000000) to sensitive sites

0

public AI-browser artifact rules in SigmaHQ, Elastic or Splunk, as of 20 Sep 2026 [15]

Close the coverage gap. Manage the attribution gap.

Thank you.

The KAPE targets, Velociraptor artifact, extractor and ES|QL hunts are in the repo. Take them, run them, tell me what you see.

github.com/sakshamtushar/AI-browser-blind-spot



Scan for the repo

YouTube

THOR-HQ

Notes

sakshamtushar.com

Talk to me

topmate.io/saksham_tushar

FOR QUESTIONS

Backup

Product timeline, Comet's install footprint, the decrypt comparison, and all sources.

One product launched and died in the time it took to plan this talk.

2 Oct 2025	Perplexity makes Comet free for everyone [10]
21 Oct 2025	OpenAI launches ChatGPT Atlas on macOS [6]
Jul 2026	OpenAI ships the official ChatGPT Chrome extension: it reads and acts on sites you're signed into [7]
9 Jul 2026	OpenAI announces Atlas is shutting down [6]
9 Aug 2026	Atlas shut down, 292 days after launch [6]
26 Aug 2026	Claude in Chrome generally available on every paid plan [8]

Windows: what to collect

Comet	%LOCALAPPDATA%\Perplexity\Comet\User Data\<profile>\	
IndexedDB\http	Prompts, thread IDs, pasted text	History · Web Data · Sessions
ps_www.perple		
city.ai or	Pages visited, ..\Local State · Network\Cookies	Keys. v20: decrypt live only
agent-filled forms		
Extensions\	The 3 bundled extensions	C:\Program Files (x86)\Perplexity\CometUpdater\updater_history.jsonl
Install and update timeline	Brave Leo	
	%LOCALAPPDATA%\BraveSoftware\Brave-Browser\User Data\<profile>\	
		AIChat
Chat timeline and model; text with the key	..\Local State · Preferences	v10 key, plaintext memories
Chrome	...\Chrome\User Data\<profile>\	
Local Extension Settings\ Gemini's own <ext-id>\ cookies and storage	Claude grants, ChatGPT heartbeat	Storage\ext\glic\
Backup · Windows paths		

macOS: what to collect

Comet	~/Library/Application Support/Comet/<profile>/	Same stores as Windows
Keychain item "Comet Safe Storage"	The decryption key; no App-Bound	Chrome
~/Library/Application Support/Google/Chrome/		<profile>/Local Extension Settings/<ext-id>/
Claude grants, ChatGPT	<profile>/Storage/ext/glic/	Gemini's own cookies and storage
NativeMessagingHosts/	Which desktop agents talk to the browser	System
~/Library/Application Support/com.apple.TCC/TCC.db	What each agent was granted	~/Library/LaunchAgents/
Leftover updates of deleted browsers	Agents	~/.claude/ · ~/.codex/ · ~/Library/Application Support/Claude/
The desktop agents' own sessions		
The bridges and logs	The bridges and permission grants tell you which agents could act, before you open a single chat store.	

Our Comet install put binaries system-wide and user data per-user.

Binaries in Program Files\Perplexity\Comet from the machine-level installer we used. A per-user installer also exists.

User data in %LOCALAPPDATA%\Perplexity\Comet\User Data.

3 machine services: two auto-updaters and an on-demand elevation service that guards encrypted cookies.

Our Brave install, by contrast, was per-user with no services.

The updater keeps its own log

`CometUpdater\updater_history.jsonl`

Every updater start and stop, with command line and parent PID. Update requests with machine details. Timestamps throughout.

Brave opens from a disk image. Comet did not.

Brave 153

v10, key protected only by the user's DPAPI

Decrypted

Comet 152

v20 App-Bound, 79 cookie rows

0 of 79

```
Comet · decrypt, live on the Windows VM
```

```
> python chromium_decrypt.py "...\\Comet\\User Data" "Profile 1"
app_bound_encrypted_key present: True
v10 AES key recovered via user-DPAPI: 32 bytes
.perplexity.ai · comet_browser · [v20-APP-BOUND, needs elevation service]
www.perplexity.ai · __Secure-next-auth.session-token · [v20-APP-BOUND]
totals: 79 cookies, v10-decrypt=0, v20-appbound=79
```

Live capture · thorhq-windows, as the user · 26 Sep 2026 · 2 of 79 rows shown

Sources, 1 of 2

-
- [1] **OX Security**, [Malicious Chrome extensions steal ChatGPT and DeepSeek conversations](#), 30 Dec 2025

 - [2] **BleepingComputer**, [Anthropic warns infostealer malware is hijacking Claude sessions](#) (quotes Anthropic's email to affected users), 30 Aug 2026

 - [3] **LayerX**, [Enterprise Browser Extension Security Report 2026](#), 9 Apr 2026

 - [4] **Chrome Web Store**, listings for [Claude](#), [ChatGPT](#), [Sider](#), [Monica](#) (user counts), read 26 Sep 2026

 - [5] **Wang, Shafiq, Vekaria (UC Davis)**, [FP-Agent: Fingerprinting AI Browsing Agents](#), arXiv:2605.01247, 2 May 2026

 - [6] **TechCrunch**, [OpenAI is shutting down Atlas](#), 9 Jul 2026

 - [7] **OpenAI**, [ChatGPT Chrome extension documentation](#), 2026

 - [8] **Anthropic**, [Claude in Chrome is now generally available](#), 26 Aug 2026
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Sources, 2 of 2

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- [9] **Google**, [Chrome AI innovations: Gemini in Chrome](#), 2026
 - [10] **TechCrunch**, [Perplexity's Comet AI browser now free](#), 2 Oct 2025
 - [11] **Perplexity Help Center**, [Comet Assistant privacy and data use](#), 2026
 - [12] **Google**, [Gemini Apps activity: retention](#), 2026
 - [13] **Microsoft**, [Conversation history in Microsoft Copilot](#), 2026
 - [14] **EricZimmerman/KapeFiles**, [Targets/Browsers \(31 targets\)](#), read 26 Sep 2026
 - [15] **Our code search** of [SigmaHQ](#), [Elastic detection-rules](#) and [Splunk security_content](#) for comet, perplexity, atlas, glic, AIChat, claude in chrome, 20 Sep 2026
 - [16] **Origin Technology**, [Claude for Chrome takeover](#): plaintext tokens and an injectable permissionStorage, Feb 2026
 - [17] **Chromium**, [ui/base/page_transition_types.h](#): core transition types and qualifiers, incl. FROM_ADDRESS_BAR and FROM_API, read 7 Oct 2026
-

Everything else is from our own lab captures: a Windows 11 VM, Elastic, and a Mac read-only, Sep–Oct 2026.